

Experimental Setup

Can an explicit discrete surface inverse and independently refined calibration resolve the unfitted fourth-order response on two incident spectra?

Experiment: gross.reception-transfer.v1 Model: signal-space.ss-ocf-1.flat-neutral-clock.v1
Runtime configuration: fixtures/research/gross-test-07-transfer.json
Plan lock: 12fa3a66a48d90185d499eee95a457d652b70ee2af87b4cbc3a0b6bb4268cbaf

MODEL AND NUMERICAL SETUP

Action: Unchanged SS OCF 1 radial Hamiltonian; exact formal neutral A,A3 and core/clock A0,A2,A4 coefficients, including quadrature logarithm square term.

Units: $c=\hbar=m=1$; cycles = phase/(2 π).

Domain: Regular origin, R=80 and domain control R=100; $h=.1,.05,.025,.0125$. Independent BVP/Newton core and independent clock eigenmode at fixed $\omega_Q=.9$. Readout $\omega_{\chi}=.41274991$ stays fixed.

Boundaries: Zero radial fields at outer boundary, T=60 before any outer reflection returns. Local $r=.1$; surface $r=14$.

Initial Data: b compact in known [18,26], $v=D_h b$. Two shapes unknown to inverse: $\cos^4$ envelope center22 width3.25 and width2.5 times $\cos(1.4(r-22))$. $A=.008, +/- .012$, quiet receiver. Clock amplitude .001 and phase0.

Solver: RK4 plus exact adjoint of the discrete acquisition propagator. Surface b and b_t every .025 through t16. SVD inverse cutoffle-10 andle-8 control. Predictors receive support/incoming rule but not shape. Output .025 with decimated .05 control. Nonlinear receivers follow all locked forecasts.

Tolerances: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\max$ sampling interval change+2e-11 cycles. Require $B\leq|P4-P2|/4$. See protocol for marker, convergence and decision rules.

COMPETING PREDICTIONS

controlled: Discrete inversion removes coarse propagation mismatch; a converged fourth-order correction improves both spectra by more than the complete numerical budget.

inadequate: A persistent resolved phase/timing residual rejects the extension; insufficient numerical resolution leaves it unresolved.

CONTROLS

frozen: Preserve action and calibration identity; method: Independently resolve the same ω_Q branch and mode without fitting ω_{χ} ; prior interpolated fine profile is a labeled diagnostic.

numeric: Bound mesh, time, box, sampling, and inverse error; method: Four meshes, finest half-dt, wide box, Hermite decimation, fixed alternate SVD cutoff.

spectrum: Fresh prediction across spectral content; method: Broad shell and carrier-modulated shell; signed amplitudes; no nonlinear receiver before all forecast locks.

PREREGISTERED DECISIONS

Experimental Setup

prediction-first (prediction-first)

Pass: All surface/input/jet/profile and marker/interval forecast hashes verify and all predictions precede all nonlinear receiver starts.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\max$ sampling interval change $+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

profile-refinement (profile-refinement)

Pass: Independent profiles have discrete residual $<2e-10$ and finest eigenvalue change ≤ 0.6 times previous change.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\max$ sampling interval change $+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

surface-inverse (surface-inverse)

Pass: Every surface relative replay residual $<1e-7$; changed SVD cutoff is included in prediction uncertainty.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\max$ sampling interval change $+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

continuum-convergence (continuum-convergence)

Pass: For both spectra at $A=.012$, finest interval change ≤ 0.6 previous mesh change plus declared nonspatial error. Otherwise unresolved.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Experimental Setup

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\text{max sampling interval change}+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

order-resolution (order-resolution)

Pass: For both spectra at $A=.012$, total absolute interval budget $B \leq |P4-P2|/4$ and $|Y|>5B$. Otherwise unresolved; never use the 5% record gate to resolve $A4$.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\text{max sampling interval change}+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

fourth-order-prediction (fourth-order-prediction)

Pass: On each resolved converged spectrum, $|Y-P4| \leq B$ and $|Y-P2|>3B$; resolved violation fails, unresolved controls defer judgment.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\text{max sampling interval change}+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

marker-transfer (marker-transfer)

Pass: All cases possess first/last Hermite markers and error ≤ 0.1 . On each nominal finest spectrum error $\leq \text{direct mesh}+\text{time}+\text{domain}+\text{inverse}+\text{sampling}+1e-6$ timing budget and decreases from fine or is below that budget.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\text{max sampling interval change}+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

sign-even (sign-even)

Pass: Finest $+.012/-0.012$ marker times differ $<1e-9$ and intervals $<1e-10$ cycles for each spectrum.

Experimental Setup

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\text{max sampling interval change}+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

conservation (conservation)

Pass: All cases charge drift $<1e-5$ and energy residual $<1\%$ of incident energy.

Fail: Resolved evidence violates the locked condition; inversion or prediction-order failure fails technical inference.

Unresolved: Missing markers, unconverged sequence or B greater than one quarter of order separation defers order discrimination.

Uncertainty: Nominal $A=.012$. $B=|Y_{\text{finest}}-Y_{\text{finer}}|+|Y_{\text{finest}}-Y_{\text{time}}|+|Y_{\text{wide}}-Y_{\text{finer}}|+|P4_{\text{inverse}}-P4_{\text{finest}}|+\text{max sampling interval change}+2e-11$ cycles. Require $B \leq |P4-P2|/4$. See protocol for marker, convergence and decision rules.

VISUAL QUESTIONS

surface-transfer [evidentiary]: How does the external record determine the local neutral and clock response?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface field, local clock phase, marker times, error budgets

controlled: converged resolved response

inadequate: persistent residual or unresolved order separation

Controls: frozen, numeric, spectrum; uncertainty: Direct mesh/time/domain/inverse/sampling differences; no fitted error reduction.

Data: plot-data/summary.json, plot-data/traces.csv, plot-data/surface.csv; transforms: Matched quiet subtraction; unfitted quadrature expansion; Hermite marker roots; direct numerical differences.

profile-convergence [evidentiary]: Does independent radial calibration converge beyond the interpolated old profile?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface field, local clock phase, marker times, error budgets

controlled: converged resolved response

inadequate: persistent residual or unresolved order separation

Controls: frozen, numeric, spectrum; uncertainty: Direct mesh/time/domain/inverse/sampling differences; no fitted error reduction.

Data: plot-data/summary.json, plot-data/traces.csv, plot-data/profiles.csv; transforms: Matched quiet subtraction; unfitted quadrature expansion; Hermite marker roots; direct numerical differences.

Experimental Setup

order-budget [evidentiary]: Is the fourth-order improvement larger than every declared numerical uncertainty?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface field, local clock phase, marker times, error budgets

controlled: converged resolved response

inadequate: persistent residual or unresolved order separation

Controls: frozen, numeric, spectrum; uncertainty: Direct mesh/time/domain/inverse/sampling differences; no fitted error reduction.

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Matched quiet subtraction; unfitted quadrature expansion; Hermite marker roots; direct numerical differences.

marker-convergence [evidentiary]: Do first and last marker prediction residuals decrease under the controls?

View: PNG/SVG/PDF with exact CSV/JSON; observables: surface field, local clock phase, marker times, error budgets

controlled: converged resolved response

inadequate: persistent residual or unresolved order separation

Controls: frozen, numeric, spectrum; uncertainty: Direct mesh/time/domain/inverse/sampling differences; no fitted error reduction.

Data: plot-data/summary.json, plot-data/traces.csv; transforms: Matched quiet subtraction; unfitted quadrature expansion; Hermite marker roots; direct numerical differences.

RESOURCE LIMITS

```
{"max_memory_mb": 2048, "max_output_mb": 300, "max_wall_seconds": 7600}
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PROHIBITED INFERENCES

Inspected prior Test 7 data cannot support a fresh prediction claim.

Test 7 original scientific fail remains unchanged; Test 8 is blocked until separate acceptance.

No two-object recoil, invariance, gravity, angular stability or emergent spacetime inference.